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Aircraft nacelle equipped with at least one thrust reversal device comprising at least one longitudinal deflector configured to house a longitudinal actuator

Abstract

An aircraft nacelle equipped with a thrust reversal device, which comprises: at least one longitudinal actuator for activating or deactivating the thrust reversal device, at least one deflection system configured to deflect an air stream channeled in the nacelle towards a lateral opening in the nacelle, in the activated state, at least one orientation system having: at least one transverse deflector configured to orient the air stream deflected by the deflection system towards the upstream end of the nacelle, for each longitudinal actuator, a hollow longitudinal deflector, which is configured to house the longitudinal actuator and has an aerodynamic profile that has a leading edge oriented towards the deflection system.

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Background/Summary

CROSS-REFERENCES TO RELATED APPLICATIONS

(1) This application claims the benefit of French Patent Application Number 2306154 filed on Jun. 16, 2023, the entire disclosure of which is incorporated herein by way of reference.

FIELD OF THE INVENTION

(2) The present application relates to an aircraft nacelle equipped with at least one thrust reversal device comprising a longitudinal deflector configured to house a longitudinal actuator and to an aircraft comprising at least one aircraft propulsion unit having such a nacelle.

BACKGROUND OF THE INVENTION

(3) According to one embodiment shown in FIGS. **1** to **3**, an aircraft **10** comprises a plurality of propulsion units **12** positioned beneath each of the wings **14** of the aircraft **10** and connected to the latter by pylons **16**. Each propulsion unit **12** comprises a powerplant **18** positioned inside a nacelle **20**. The powerplant **18** comprises a fan that has an axis of rotation **A18**.

(4) For the remainder of the description, a longitudinal direction is parallel to the axis of rotation **A18**. A longitudinal plane is a plane containing the axis of rotation **A18** and a transverse plane is a plane perpendicular to the axis of rotation **A18**. The concepts of front/upstream and rear/downstream refer to the direction of flow of an air stream in the nacelle **20**, this stream flowing from the front (from upstream) to the rear (downstream).

(5) The nacelle **20** has an approximately tubular shape and delimits, with the powerplant **18**, an annular duct **22** in which a secondary air stream flows. The nacelle **20** comprises an upstream part **24** and a downstream part **26**, which extends to a trailing edge **28**.

(6) The nacelle **20** comprises a thrust reversal device **30** positioned in the downstream part **26** and configured to occupy an activated state, in which it deflects, outwards and upstream of the nacelle **20**, at least some of the secondary air stream circulating in the annular duct **22**, and an inactivated state, in which it does not deflect the secondary air stream circulating in the annular duct **22**.

(7) The thrust reversal device **30** comprises at least one lateral opening **32** (shown in FIG. **3**) passing through the nacelle **20**. According to one configuration, the downstream part **26** of the nacelle **20** is movable between a first position, corresponding to the inactivated state of the thrust reversal device **30**, in which the downstream part **26** is in contact with the upstream part **24** (as illustrated in FIG. **2**), and a second position, corresponding to the activated state of the thrust reversal device **30**, in which the downstream part **26** is moved away from the upstream part **24** so as to create the lateral opening **32** (as illustrated in FIG. **3**).

(8) In order to move the downstream part **26**, the thrust reversal device **30** comprises a plurality of longitudinal actuators **34**, which connect the upstream and downstream parts **24**, **26**, are distributed over the circumference of the nacelle and are configured to move the downstream part **26** from the front to the rear, or vice versa, with respect to the upstream part **24**.

(9) The thrust reversal device **30** comprises a deflection system **36** configured to occupy a retracted position (shown in FIG. **2**), when the thrust reversal device **30** is in the inactivated state, in which the deflection system **36** does not interfere with the secondary air stream, and a deployed position (shown in FIG. **3**), when the thrust reversal device **30** is in the activated state, in which the deflection system **36** interferes with the secondary air stream and orients it towards the lateral opening **32**.

(10) The thrust reversal device **30** also comprises an orientation system **38** configured to orient, outwards and upstream of the nacelle, the air stream deflected by the deflection system **36**. In addition, the downstream part **26** comprises a cavity **40**, which is open in the direction of the upstream part **24** and is configured to house the orientation system **38** and the longitudinal actuators **34**. This cavity **40** is delimited by inner and outer walls **26.1**, **26.2**, which are spaced apart in the direction of the upstream part **24** and connected at the trailing edge **28**.

(11) According to one embodiment shown in FIG. **4**, the orientation system **38** comprises a plurality of cascades of vanes **42**, which are positioned at the lateral opening **32** and each have first walls positioned in approximately longitudinal planes and second walls positioned in approximately transverse planes. The second walls take the form of a vane in order to deflect upstream the air stream leaving the lateral opening **32**. Some of these cascades of vanes **42** are spaced apart in order to house the longitudinal actuators **34**.

(12) This embodiment is not satisfactory because the cascades of vanes **42** have a significant mass. Furthermore, the fact that the regions occupied by the longitudinal actuators **34** do not have an orientation system **38** impairs the effectiveness of the thrust reversal device **30**.

SUMMARY OF THE INVENTION

(13) The present invention aims to overcome all or some of the aforementioned drawbacks.

(14) To this end, the invention relates to an aircraft nacelle, which has an upstream end and a downstream end and is configured to channel an air stream flowing in a longitudinal direction from the upstream end to the downstream end, the nacelle having at least one thrust reversal device, which comprises: at least one lateral opening, which is remote from the upstream and downstream ends, passes through the nacelle and is delimited by upper and lower edges, at least one movable part configured to occupy a first position, corresponding to an inactivated state of the thrust reversal device, in which the movable part shuts off the lateral opening, and a second position, corresponding to an activated state of the thrust reversal device, in which the movable part at least partially clears the lateral opening, at least one longitudinal actuator, which passes through the lateral opening and is configured to move the movable part, at least one deflection system configured to occupy a first position, when the thrust reversal device is in the inactivated state, and a second position, when the thrust reversal device is in the activated state, in which the deflection system interferes with the air stream channeled in the nacelle and orients it towards the lateral opening, at least one orientation system configured to orient the air stream deflected by the deflection system.

(15) According to the invention, the orientation system comprises: at least one transverse deflector positioned approximately in a plane perpendicular to the longitudinal direction and configured to orient the air stream deflected by the deflection system towards the upstream end of the nacelle, each deflector having first and second ends respectively connected to a support at the upper and lower edges of the lateral opening, for each longitudinal actuator, a hollow longitudinal deflector, which is configured to house the longitudinal actuator and has an aerodynamic profile that has a leading edge oriented towards the deflection system.

(16) This solution makes it possible to limit the impact of the longitudinal actuators on the effectiveness of the thrust reversal device.

(17) According to another feature, each longitudinal deflector is connected to each transverse deflector.

(18) According to another feature, each transverse deflector comprises, for each longitudinal actuator passing through it, a cutout following the profile of the longitudinal deflector in which the longitudinal actuator is housed.

(19) According to another feature, each longitudinal deflector comprises first and second walls, each of the first and second walls having an inner edge and an outer edge that is further away from the deflection system than the inner edge is, the inner edges of the first and second walls being contiguous so as to form the leading edge, the outer edges being spaced apart.

(20) According to another feature, the longitudinal deflector comprises a third wall connecting the outer edges of the first and second walls.

(21) According to another feature, the first wall has a first part, which extends from the inner edge and has a first radius of curvature, and a second part, which extends from the outer edge to the first part and has a second radius of curvature. In addition, the second wall comprises a third part, which extends from the inner edge and has a third radius of curvature, and a fourth part, which extends from the outer edge to the third part and has a fourth radius of curvature.

(22) According to other features, the third and fourth radii of curvature are equal and/or the second and fourth radii of curvature are equal and/or the first radius of curvature is significantly smaller than the third radius of curvature.

(23) According to another feature, the orientation system comprises a plurality of transverse deflectors each having inner and outer edges.

(24) According to another feature, the inner edges of the first and second walls of each longitudinal deflector are situated at at least one of the inner edges of the transverse deflectors.

(25) According to another feature, the outer edges of the first and second walls of each longitudinal deflector are situated at the outer edge of at least one of the transverse deflectors.

- (26) According to another feature, the first and second walls have geometries that are determined on the basis of an orientation desired for the air stream leaving the lateral opening.
- (27) According to another feature, each longitudinal deflector extends between first and second ends, its aerodynamic profile varying between the first and second ends.
- (28) The invention also relates to an aircraft comprising at least one nacelle according to one of the preceding features.
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Description

BRIEF DESCRIPTION OF THE DRAWINGS

- (1) Other features and advantages will become apparent from the following description of the invention, the description being given solely by way of example and with reference to the appended drawings, in which:
- (2) FIG. 1 is a perspective view of an aircraft and a propulsion unit,
- (3) FIG. 2 is a schematic section through a propulsion unit equipped with a thrust reversal device in the inactivated state, illustrating an embodiment of the prior art,
- (4) FIG. 3 is a schematic section through the propulsion unit shown in FIG. 2, the thrust reversal device being in the activated state,
- (5) FIG. 4 is a perspective view of a part of a thrust reversal device, illustrating an embodiment of the prior art,
- (6) FIG. 5 is a side view of a part of a propulsion unit having a thrust reversal device in the activated state, illustrating an embodiment of the invention,
- (7) FIG. 6 is a schematic section through a propulsion unit equipped with a thrust reversal device in the inactivated state, illustrating an embodiment of the invention,
- (8) FIG. 7 is a schematic section through the propulsion unit shown in FIG. 6, the thrust reversal device being in the activated state,
- (9) FIG. 8 is a perspective view of a part of a longitudinal deflector configured to house a longitudinal actuator, illustrating an embodiment of the invention,
- (10) FIG. 9 is a perspective view, from the outside of the nacelle, of a part of an orientation system of a thrust reversal device, illustrating an embodiment of the invention,
- (11) FIG. 10 is a perspective view, from the inside of the nacelle, of a part of an orientation system of a thrust reversal device, illustrating an embodiment of the invention,
- (12) FIG. 11 is a longitudinal section through a part of an orientation system of a thrust reversal device, illustrating an embodiment of the invention,
- (13) FIG. 12 is a cross section through a part of the orientation system shown in FIG. 11.

DETAILED DESCRIPTION OF THE PREFERRED EMBODIMENTS

- (14) According to an embodiment shown in FIGS. 5, 6 and 7, a propulsion unit **50** comprises a powerplant **52** positioned inside a nacelle **54**. The latter has an approximately tubular shape and delimits, with the powerplant **52**, an annular duct **56** in which a secondary air stream flows. The nacelle **54** comprises an upstream part **54.1** and a downstream part **54.2**, which has a trailing edge **58**. The nacelle **54** has an outer surface **F54**.
- (15) According to one application, an aircraft comprises at least one such propulsion unit. Irrespective of the embodiment, the nacelle **54** is tubular and extends between upstream and downstream ends. The nacelle **54** is configured to channel an air stream **60** in a longitudinal direction, from the upstream end to the downstream end.
- (16) The nacelle **54** comprises at least one thrust reversal device **62** positioned in the downstream part **54.2** and configured to occupy an activated state (shown in FIGS. 5 and 7), in which it deflects, outwards and towards the upstream end of the nacelle **54**, at least some of the air stream **60** circulating in the annular duct **56**, and an inactivated state (shown in FIG. 6), in which it does

not deflect the air stream **60** circulating in the annular duct **56**.

(17) The thrust reversal device **62** comprises at least one lateral opening **64**, which passes through the nacelle **54**, is remote from the upstream and downstream ends of the nacelle **54**, and is configured to place the annular duct **56** and the outside of the nacelle **54**, notably its outer surface **F54**, in communication, and at least one movable part **66** configured to occupy a first position, corresponding to the inactivated state of the thrust reversal device **62** (shown in FIG. 6), in which the movable part **66** shuts off the lateral opening **64**, and a second position, corresponding to the activated state of the thrust reversal device **62** (shown in FIGS. 5 and 7), in which the movable part **66** at least partially clears the lateral opening **64**. This lateral opening **64** is delimited by upper and lower edges.

(18) According to one embodiment, the nacelle **54** comprises two lateral openings **64**, which are substantially symmetrical with respect to a vertical longitudinal plane passing through the axis of the powerplant.

(19) According to one configuration, the downstream part **54.2** is a movable O-shaped tubular cowl, forming the movable part **66**, which moves in translation in the longitudinal direction between a closed position (shown in FIG. 6), in which the downstream part **54.2** is in contact with the upstream part **54.1**, when the thrust reversal device **62** is in the inactivated state, and an open position (shown in FIGS. 5 and 7), in which the downstream part **54.2** is moved away from the upstream part **54.1** so as to create the one or more lateral openings **64** between the upstream and downstream parts **54.1**, **54.2** when the thrust reversal device **62** is in the activated state.

(20) The downstream part **54.2** comprises an inner wall **68.1** and an outer wall **68.2**, which are connected at the trailing edge **58**, the inner and outer walls **68.1**, **68.2** delimiting between them a cavity **70** that is open in the direction of the upstream part **54.1**.

(21) Of course, the invention is not limited to this configuration for the movable part **66**.

(22) The thrust reversal device **62** comprises at least one deflection system **72** configured to occupy a retracted position (shown in FIG. 6), when the thrust reversal device **62** is in the inactivated state, in which the deflection system **72** does not interfere with the air stream **60** channeled by the nacelle **54**, and a deployed position (shown in FIG. 7), when the thrust reversal device **62** is in the activated state, in which the deflection system **72** is deployed in the annular duct **56**, interferes with the air stream **60** and orients it in radial directions (which are perpendicular to the longitudinal direction) towards the lateral opening **64**.

(23) According to one embodiment, for each lateral opening **64**, the deflection system **72** comprises at least one panel **72.1** and, for each panel **72.1**, at least one articulation **72.2** connecting the panel **72.1** to the nacelle **54** and at least one link rod **72.3** connecting the panel **72.1** to the powerplant **52**. Of course, the invention is not limited to this embodiment for the deflection system **72**. As a result, each panel could be connected to the powerplant **52** via an articulation.

(24) The thrust reversal device **62** also comprises at least one orientation system **74** configured to orient at least some of the air stream **60** deflected by the deflection system **72** and leaving via the lateral opening **64**. According to one arrangement, the orientation system **74** is positioned at the lateral opening **64**.

(25) According to one arrangement, the orientation system **74** and the cavity **70** are configured such that, in the inactivated state, the orientation system **74** is positioned in the cavity **70**.

(26) According to one embodiment, the orientation system **74** comprises at least one transverse deflector **76**, which is positioned approximately in a transverse plane and extends between first and second ends **76.1**, **76.2**. Each transverse deflector **76** has inner and outer edges **76.3**, **76.4** that are curved, substantially in the shape of a circular arc, and connect the first and second ends **76.1**, **76.2**, the inner edge **76.3** being closer to the deflection system **72** than the outer edge **76.4** is. According to one configuration, the inner and outer edges **76.3**, **76.4** have curvatures adapted to those of the inner and outer walls **68.1**, **68.2** of the downstream part **54.2** of the nacelle **54** such that the orientation system **74** can be housed in the cavity **70** when the thrust reversal device **62** is in the

inactivated state.

(27) Each transverse deflector **76** is configured to orient, in the radial directions and towards the upstream end, the air stream deflected by the deflection system **72** and leaving the lateral opening **64**.

(28) According to one embodiment, each transverse deflector **76** comprises a blade, in the shape of a ring portion, which has a concave curvature (the center of the radius of curvature being offset upstream with respect to the transverse deflector **76**) in longitudinal planes.

(29) According to one configuration, each transverse deflector **76** has a width (distance separating the inner and outer edges **76.3**, **76.4**) and a curvature that are constant between its first and second ends **76.1**, **76.2**. As a result, the air stream leaving the nacelle **54** is substantially constant around the perimeter of the nacelle **54**. According to one arrangement, the transverse deflectors **76** have different widths that decrease from upstream to downstream, as illustrated in FIG. **11**.

(30) The number of transverse deflectors **76** and their shape are determined so as to obtain the same aerodynamic function as the cascades of the prior art, specifically a sufficient flow rate to avoid operational problems with the fan of the powerplant **52** and effectiveness as regards reverse thrust.

(31) According to one embodiment, the orientation system **74** comprises at least one support **78**, which is secured to the nacelle **54** and to which each transverse deflector **76** is connected.

(32) According to an embodiment shown in FIG. **5**, the support **78** comprises first and second uprights **80.1**, **80.2**, which are parallel to one another and to the longitudinal direction and are positioned above and below the lateral opening **64**, the first end **76.1** of each transverse deflector **76** being connected to the first upright **80.1**, the second end **76.2** of each transverse deflector **76** being connected to the second upright **80.2**. Irrespective of the embodiment, the first and second ends **76.1**, **76.2** of each transverse deflector **76** are respectively connected to the support **78** at the upper and lower edges of the lateral opening **64**.

(33) According to one configuration, the support **78** comprises an upstream crosspiece **82**, which is oriented in a direction parallel to a transverse plane, is positioned upstream of the lateral opening **64**, and to which the first and second uprights **80.1**, **80.2** are connected. The upstream crosspiece **82** has a curvature that is substantially identical to that of the nacelle **54**. The upstream crosspiece **82** is secured to the upstream part **54.1** of the nacelle **54** or integrated therein.

(34) The support **78** comprises a downstream crosspiece **84**, which is oriented in a direction parallel to a transverse plane and positioned downstream of the lateral opening **64**. The downstream crosspiece **84** has a curvature that is substantially identical to that of the nacelle **54**.

(35) According to one configuration, the upstream and downstream crosspieces **82**, **84** are parallel to one another and connected by the first and second uprights **80.1**, **80.2**. As a result, the upstream and downstream crosspieces **82**, **84** and the first and second uprights **80.1**, **80.2** form a frame delimiting the lateral opening **64**.

(36) The thrust reversal device **62** comprises at least one longitudinal actuator **86**, which passes through the lateral opening **64** and is configured to move the movable part **66** and to occupy a retracted state, corresponding to the inactivated state of the thrust reversal device **62**, in which the movable part **66** shuts off the lateral opening **64**, and a deployed state, corresponding to the activated state of the thrust reversal device **62**, in which the movable part **66** at least partially clears the lateral opening **64**. Each longitudinal actuator **86** is oriented in a direction that is substantially parallel to the longitudinal direction.

(37) According to one configuration, each longitudinal actuator **86** is a cylinder and connects the upstream and downstream parts **54.1**, **54.2**.

(38) According to one arrangement, the thrust reversal device **62** comprises a plurality of longitudinal actuators **86**, for example four longitudinal actuators **86**, which are distributed over the circumference of the nacelle **54** and positioned symmetrically with respect to the vertical longitudinal plane.

(39) Each longitudinal actuator **86** interferes with the transverse deflectors **76** of the orientation

system **74**. Each transverse deflector **76** comprises, for each longitudinal actuator **86** passing through it, a cutout **88** (shown in FIG. **12**).

(40) In order to limit the impact of the longitudinal actuators **86** on thrust reversal, the orientation system **74** comprises, for each longitudinal actuator **86**, a hollow longitudinal deflector **90**, which is configured to house the longitudinal actuator **86** and has an aerodynamic profile that has a leading edge **92** oriented towards the deflection system **72**. This longitudinal deflector **90** flares away from the deflection system **72**.

(41) According to one arrangement, each longitudinal deflector **90** passes all the way through the lateral opening **64** and has a first end **90.1**, which is connected to the upstream crosspiece **82**, and a second end **90.2**, which is connected to the downstream crosspiece **84**.

(42) According to one embodiment, each longitudinal deflector **90** comprises first and second walls **94**, **96** forming a V shape in a transverse plane. Each of the first and second walls **94**, **96** comprises an inner edge **94.1**, **96.1** and an outer edge **94.2**, **96.2** that faces the inner edge **94.1**, **96.1** and is further away from the deflection system **72** than the inner edge **94.1**, **96.1** is, the inner edges **94.1**, **96.1** of the first and second walls **94**, **96** being contiguous so as to form the leading edge **92**, the outer edges **94.2**, **96.2** being spaced apart so that the first and second walls **94**, **96** form a housing **98** configured to house a longitudinal actuator **86**.

(43) According to one configuration, the inner edges **94.1**, **96.1** of the first and second walls **94**, **96** and the leading edge **92** of each longitudinal deflector **90** are situated at at least one of the inner edges **76.3** of the transverse deflectors **76**. The outer edges **94.2**, **96.2** of the first and second walls **94**, **96** of each longitudinal deflector **90** are situated at the outer edge **76.4** of at least one of the transverse deflectors **76**.

(44) According to a configuration shown in FIG. **8**, the longitudinal deflector **90** comprises a third wall **100** connecting the outer edges **94.2**, **96.2** of the first and second walls so as to form a tubular longitudinal deflector **90**.

(45) According to one embodiment, each longitudinal deflector **90** is connected to each transverse deflector **76**. Connecting the longitudinal deflectors **90** to the transverse deflectors **76** contributes to stiffening the latter. According to one configuration, the cutout **88** in each transverse deflector **76** follows the profile of the longitudinal deflector **90**.

(46) According to one embodiment, the longitudinal deflector **90** has a constant cross section between its first and second ends **90.1**, **90.2**. According to another embodiment, the longitudinal deflector **90** has a non-constant cross section that varies between its first and second ends **90.1**, **90.2**, the cross section varying on the basis of the orientation desired for the air stream leaving the lateral opening **64**.

(47) According to one embodiment, the longitudinal deflector **90** has a cross section that is symmetrical with respect to a longitudinal plane, as illustrated in FIG. **8**. According to another embodiment, the longitudinal deflector has an asymmetrical cross section. By way of example, one of the first and second walls **94**, **96** has a convex profile and the other has a profile comprising a convex portion and a concave portion.

(48) According to one embodiment, the first and second walls **94**, **96** are flat. According to another embodiment, at least one of the first and second walls **94**, **96** has a curved profile.

(49) According to one configuration, the first and second walls **94**, **96** may have geometries that are determined on the basis of the orientation desired for the air stream leaving the lateral opening **64**. As a result, the longitudinal deflector **90** has an aerodynamic profile that varies between the first and second ends **90.1**, **90.2** on the basis of the orientation desired for the air stream leaving the lateral opening **64**.

(50) According to one embodiment shown in FIG. **12**, the first wall **94** has a first part **P1**, which extends from the inner edge **94.1** and has a first radius of curvature, and a second part **P2**, which extends from the outer edge **94.2** to the first part **P1** and has a second radius of curvature. The second wall **96** comprises a third part **P3**, which extends from the inner edge **96.1** and has a third

radius of curvature, and a fourth part P4, which extends from the outer edge 96.2 to the third part P3 and has a fourth radius of curvature. According to one configuration, the third and fourth radii of curvature may be equal. The second and fourth radii of curvature may be equal. The first radius of curvature is significantly (i.e., at least 10%) smaller than the third radius of curvature.

(51) Of course, the invention is not limited to this geometry for the longitudinal deflectors 90.

(52) While at least one exemplary embodiment of the present invention(s) is disclosed herein, it should be understood that modifications, substitutions and alternatives may be apparent to one of ordinary skill in the art and can be made without departing from the scope of this disclosure. This disclosure is intended to cover any adaptations or variations of the exemplary embodiment(s). In addition, in this disclosure, the terms “comprise” or “comprising” do not exclude other elements or steps, the terms “a” or “one” do not exclude a plural number, and the term “or” means either or both. Furthermore, characteristics or steps which have been described may also be used in combination with other characteristics or steps and in any order unless the disclosure or context suggests otherwise. This disclosure hereby incorporates by reference the complete disclosure of any patent or application from which it claims benefit or priority.

Claims

1. An aircraft nacelle comprising: an upstream end and a downstream end, the aircraft nacelle configured to channel an air stream flowing in a longitudinal direction from the upstream end to the downstream end; and at least one thrust reversal device which comprises: at least one lateral opening, which is remote from the upstream and downstream ends, which passes through the aircraft nacelle, and which is delimited by upper and lower edges, at least one movable part configured to occupy a first position, corresponding to an inactivated state of the at least one thrust reversal device, in which the at least one movable part shuts off the at least one lateral opening, and a second position, corresponding to an activated state of the at least one thrust reversal device, in which the at least one movable part at least partially clears the at least one lateral opening, at least one longitudinal actuator, which passes through the at least one lateral opening and is configured to move the at least one movable part, at least one deflection system configured to occupy a first position, when the at least one thrust reversal device is in the inactivated state, and a second position, when the at least one thrust reversal device is in the activated state, in which the at least one deflection system interferes with the air stream channeled in the aircraft nacelle and orients the air stream towards the at least one lateral opening, at least one orientation system configured to orient the air stream deflected by the at least one deflection system, wherein the at least one orientation system comprises: at least one transverse deflector positioned approximately in a plane perpendicular to the longitudinal direction and configured to orient the air stream deflected by the at least one deflection system towards the upstream end of the aircraft nacelle, each transverse deflector having first and second ends respectively connected to a support at the upper and lower edges of the at least one lateral opening, and, for each longitudinal actuator, a hollow longitudinal deflector, which is configured to house the longitudinal actuator and has an aerodynamic profile that has a leading edge oriented towards the at least one deflection system; wherein each longitudinal deflector is connected to each transverse deflector and comprises first and second walls; and wherein at least one of the first and second walls of the longitudinal deflector has a curved profile; and wherein each of the first and second walls has an inner edge and an outer edge that is further away from the at least one deflection system than the inner edge is, the inner edges of the first and second walls being contiguous so as to form the leading edge, the outer edges being spaced apart, wherein the first and second walls of each longitudinal deflector form a V shape at the leading edge in a transverse plane.

2. The aircraft nacelle according to claim 1, wherein each transverse deflector comprises, for each longitudinal actuator passing therethrough, a cutout following the profile of the longitudinal

deflector in which the longitudinal actuator is housed.

3. The aircraft nacelle according to claim 1, wherein each longitudinal deflector comprises a third wall connecting the outer edges of the first and second walls.

4. The aircraft nacelle according to claim 1, wherein the first wall has a first part, which extends from the inner edge and has a first radius of curvature, and a second part, which extends from the outer edge to the first part and has a second radius of curvature, and wherein the second wall comprises a third part, which extends from the inner edge and has a third radius of curvature, and a fourth part, which extends from the outer edge to the third part and has a fourth radius of curvature.

5. The aircraft nacelle according to claim 4, wherein the third and fourth radii of curvature are equal, or the second and fourth radii of curvature are equal, or the first radius of curvature is significantly smaller than the third radius of curvature, or a combination thereof.

6. The aircraft nacelle according to claim 1, wherein the at least one orientation system having at least one transverse deflector comprises a plurality of transverse deflectors each having inner and outer edges and in that the inner edges of the first and second walls of each longitudinal deflector are situated least one of the inner edges of the transverse deflectors from the plurality of transverse deflectors.

7. The aircraft nacelle according to claim 1, wherein the at least one orientation system having at least one transverse deflector comprises a plurality of transverse deflectors each having inner and outer edges and in that the outer edges of the first and second walls of each longitudinal deflector are situated at the outer edge of at least one of the transverse deflectors from the plurality of transverse deflectors.

8. The aircraft nacelle according to claim 1, wherein the first and second walls have geometries that are determined on a basis of an orientation desired for the air stream leaving the at least one lateral opening.

9. The aircraft nacelle according to claim 1, wherein each hollow longitudinal deflector extends between first and second ends and has an aerodynamic profile varying between the first and second ends of the hollow longitudinal deflector.

10. An aircraft comprising: the nacelle according to claim 1.

11. The aircraft nacelle according to claim 1, wherein each longitudinal deflector has a cross section that is symmetrical with respect to a longitudinal plane.

12. The aircraft nacelle according to claim 1, wherein each longitudinal deflector has a cross section that is asymmetrical with respect to a longitudinal plane.

13. The aircraft nacelle according to claim 1, wherein one of the first and second walls has a convex profile and the other has a profile comprising a convex portion and a concave portion.
